IMPLEMENTATION SPECIFICATION & API DOCUMENT

IHH Healthcare Berhad

Digital Certificate and Digital Signing Solution (DSS)

Version 1.1

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GLOSSARY

This glossary may contain context-specific technical terms, abbreviations or everyday concepts that have a special meaning in the given context.

NO.	TERMS	DESCRIPTION
1	**AP**	Authorised Personnel
2	**AATL**	Adobe Approved Trust List
3	**DSS**	Digital Signing Solution
4	**KMC**	Key Management Centre
5	**LOA**	Letter of Authorization
6	**LTV**	Long Term Validation
7	**MYCRS**	Pos Digicert Certificate Registration System
8	**OCSP**	Online Certificate Status Protocol
9	**PDF**	Portable Document Format
10	**QC**	Pos Digicert Quality Control
11	**RA**	Pos Digicert Registration Authority Department
12	**RO**	Pos Digicert Registration Officer
13	**RP**	Pos Digicert Registration Personnel
14	**SLA**	Service Level Agreement
15	**SSM**	Suruhanjaya Syarikat Malaysia
16	**System**	System that will implement PKI service
17	**TSA**	Time Stamping Authority
18	**VS**	Pos Digicert Verification Specialist

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1 Introduction

This document is the **Implementation Specification (IS)** and **API document** for **IHH Healthcare Berhad** on the implementation and integration of Digital Certificate and Digital Signing Solution (DSS). In this document, the project implementation and API information are described in detail.

This document will consist of

- **Related Information** on **page 16**
- **Pre-requisite** on **page 17**
- **Digital Certificate Request and Issuance for User** on **page 19**
- **Authorised Personnel (AP)** on **page 33**
- **Digital Signing Solution (DSS)** on **page 34**
- **Contact Us** on **page 48**

*Note: API Parameter is case sensitive.

1.1 Objective

The objectives of this document are:

- To explain in detail, the Digital Certificate and DSS implementation
- To use as base development by IHH Healthcare Berhad
- To use as solution deliverable reference by Pos Digicert

1.2 Intended Audience

This document is intended for:

- Personnel responsible for the approval of the Digital Certificate and DSS implementation specification
- Technical Lead / Developer involved in the design and implementation of Digital Certificate and DSS

1.3 Scope

List of Pos Digicert scope of work:

- To advise the implementation for Digital Certificate and DSS
- Configuration of Digital Certificate at Pos Digicert
- Installation and configuration of DSS at IHH Healthcare Berhad for **Staging** and **Production**

1.4 Assumption and Constraint

Assumption and constraint are as below,

- As for standard, System will **not save/cache** Roaming Certificate PIN. Saved/cached Roaming Certificate PIN will be under IHH full responsibility as per IHH's letter of responsibility.
- System will make sure that Identification No. (MyKad/Passport No.) And Name **remain the same as per certificate request** whenever certificate has been issued. Certificate issuance will be based on MyKad/Passport information
- TOTP function has been removed based on IHH Healthcare Berhad URS session on 15/01/2025 and will be under IHH full responsibility as per IHH's letter of responsibility.
- This document will only describe and explain the standard process of integration on API calling. Description/requirement on how System should call / handle API service will not be explained in this document. Such requirement document is up to IHH Healthcare Berhad

Please refer to the next page.

2 Solution Overview

2.1 Components Overview

Below are applicable components for project implementation.

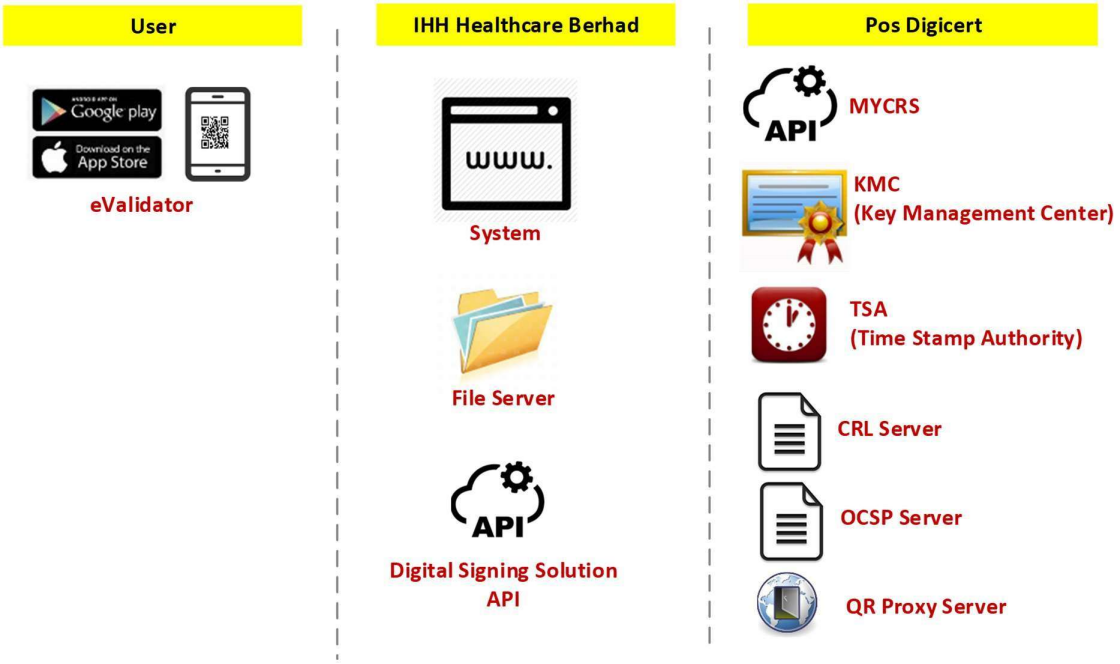


Figure 1: Components Overview

NO	ITEM	RESIDE AT	REMARK
1	**System**	**IHH Healthcare Berhad**	IHH system that will implement PKI Solution
2	**DSS Server**		Digital Signing Solution that consists of signing API and related signing API ▪ Server (Oracle Linux 8): 4 core, 16GB RAM, 1TB HDD
3	**File Server**		File server to use by System and DSS (temporary store and retrieve PDF File)
4	**MYCRS**	**Pos Digicert**	MYCRS (Certificate Registration System) for issuance and related digital certificate's action
5	**KMC**		Server that stores individual roaming digital certificate and related APIs (Key Management Centre)
6	**TSA**		Time Stamping Authority server. Times that appear in ADOBE signature properties
7	**CRL Server**		For Long Term Validation (LTV) and Revocation Checking purpose
8	**OCSP Server**		
9	**QR Proxy Server**		For QR Code
10	**eValidator**	**User**	To verify hardcopy against file in system

2.2 Integration Process Overview

Standard integration process overview process and IHH Healthcare integration process are as per below.

Figure 2 shows Certificate Issuance done by IHH Admin and **Figure 3** shows signing process done by Doctor.

IHH Admin will create digital certificate issuance first before Doctor able to do signing process.

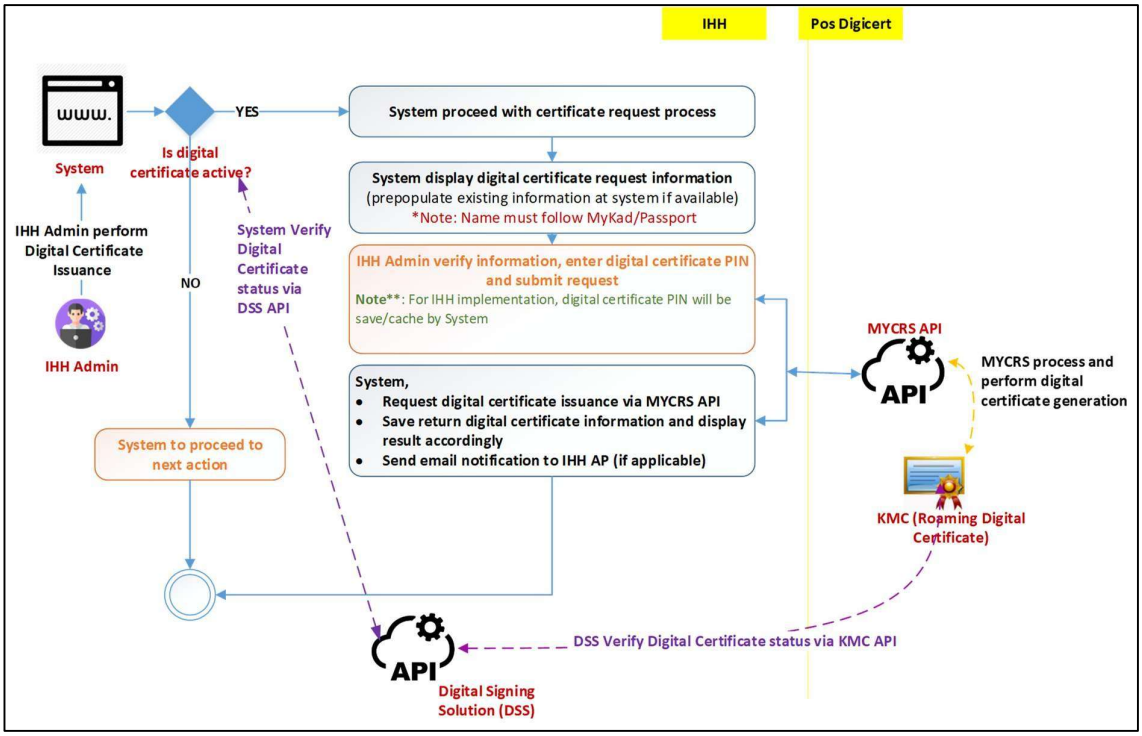


Figure 2: Certificate Issuance Integration Process Overview

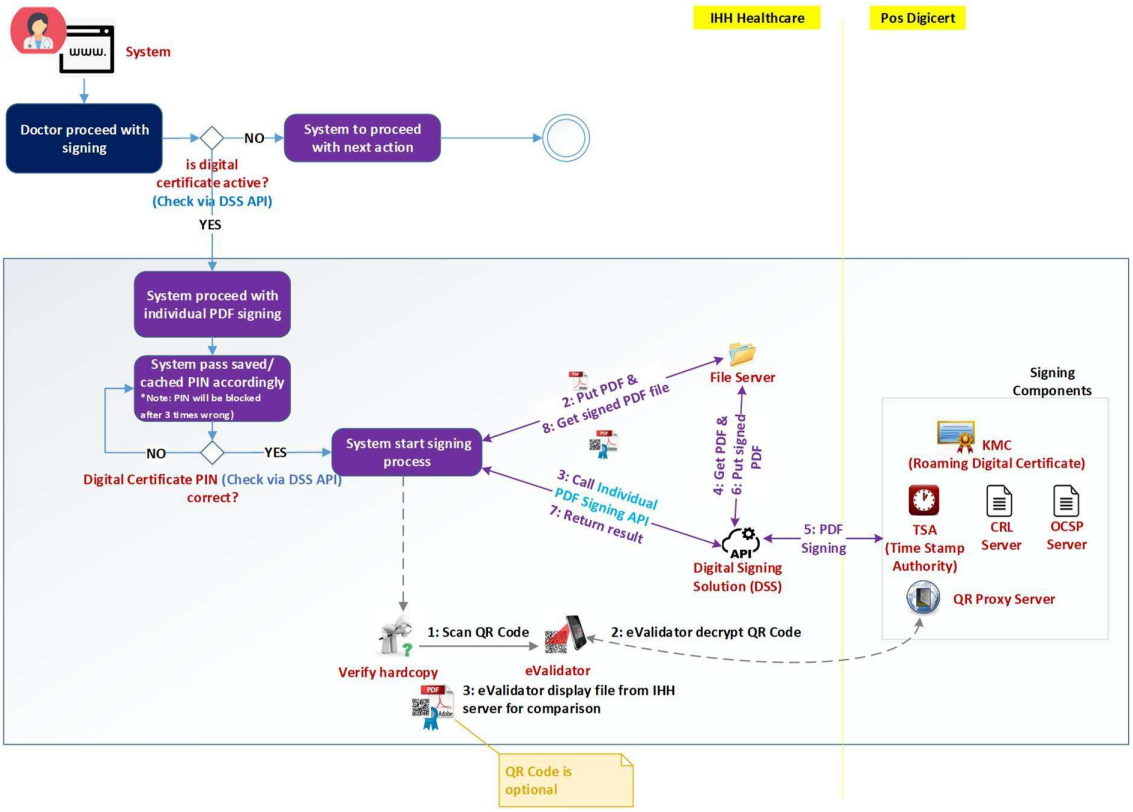
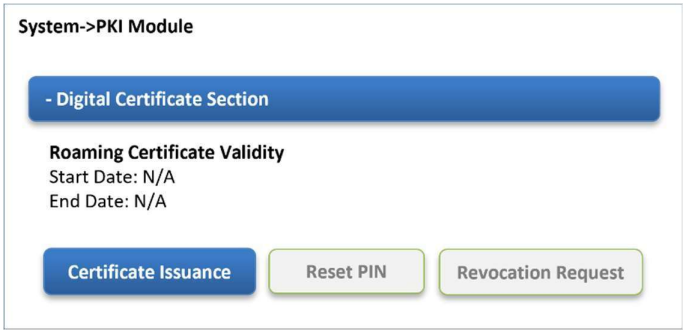


Figure 3: Signing Integration Process Overview

As per **Figure 2** and **Figure 3**, IHH Healthcare needs to implement 2 (two) main processes in system flow, that are **Digital Certificate Issuance** (is digital certificate active?) and **PDF Signing**. Please refer to details in the respective section.

2.3 Sample Standard Implementation Overview



Sample Standard Implementation by System 1: Implementation Overview

1) Allows IHH Admin to perform action regarding PKI under dedicated menu after successfully login in System
2) Control available function based on identity status and roaming digital certificate status. Example as below

Function	Logic
Digital Certificate Request & Issuance	Allows if status is expired
Reset PIN	Allows if certificate exist
Revocation Request	Allows if status is active and valid

3 Related Development Information

Related information for Development and Production environment.

3.1 Development Environment

Please refer to the below for development information.

3.1.1 Digital Certificate

Related information for Digital Certificate is below.

ITEM	VALUE	DESCRIPTION
MYCRS Project Code		
MYCRS Package Code	IHH0005-NEW	IHH New Roaming 1 Year
	IHH0002-RNW	IHH Renew Roaming 1 Year
	IHH0003-NEW	IHH New Roaming 7 Days
	IHH0004-RNW	IHH Renew Roaming 7 Days
AP Login		

3.1.2 DSS File Server

Related information for DSS is below.

*Please provide Pos Digicert the Public IP for whitelist purpose if using SAAS or Pos Digicert Staging

URL	PROTOCOL	PORT	USERNAME	PASSWORD	FILE SERVER ID	FOLDER (PUT/GET)
	SFTP	22				/home/dss/DSSFS

3.2 Production Environment

Will provide later in a separate document.

4 Pre-requisite

This section will describe the pre-requisite.

4.1 General

ID	DESCRIPTION
PREQ_GENERAL_001	Install **eValidator** mobile apps. Pos Digicert eValidator are available to be downloaded at Google Play or Apps Store

Table 1: Pre-requisite: General

4.2 System

Refer below table for the pre-requisite on implementation.

ID	DESCRIPTION
PREQ_SYSTEM_001	System must be able to invoke REST web service
PREQ_SYSTEM_002	System must make sure the existence of the PDF or signature image (if in use) file before invoking the PDF signing APIs. ***Note:** The filename for Source and Destination must be unique
PREQ_SYSTEM_003	Allows the read and write access at file server, for DSS to read the PDF file and to create the signed PDF file
PREQ_SYSTEM_004	For batch registration, the API calling queuing must be handled by System
PREQ_SYSTEM_005	For batch signing, the API calling queuing must be handled by System
PREQ_SYSTEM_006	For QR Code, • QR Code must be used during the first signing. System could not embed QR Code after the PDF file has been signed • Document validity for QR Code must be handled by System

Table 2: Pre-requisite: System

4.3 Connection

This section will describe the pre-requisite for internal and external connection.

4.3.1 Internal

Refer below table for the internal connectivity port.

ID	DESCRIPTION
PREQ_ICONN_001	Allows connectivity between System server to DSS [port 8080]
PREQ_ICONN_002	Allows connectivity between DSS to File Server [SFTP or Mapped Drive]
PREQ_ICONN_003	Allows connectivity between System server to File Server (SFTP)

Table 3: Pre-requisite: Internal Connection

4.3.2 External

Refer below table for the external connectivity port. External connection to Pos Digicert requires Pos Digicert to whitelist system IP (Applicable for staging and production).

ID	DESCRIPTION
PREQ_ECONN_001	Allows connectivity from DSS to KMC. [port 443] Production is [will provide later in a separate document] Staging is stg-kmc.posdigicert.com.my
PREQ_ECONN_002	Allows connectivity from DSS to TSA [port 443] Production is [will provide later in a separate document] Staging is trialdts.posdigicert.com.my/ihh/tsa
PREQ_ECONN_003	Allows connectivity from DSS to CRL [port 80] Production is [will provide later in a separate document] Staging is trialcrl.posdigicert.com.my or trialcrl.digicert.com.my
PREQ_ECONN_004	Allows connectivity from DSS to OCSP [port 80] Production is [will provide later in a separate document] Staging is trialocsp.posdigicert.com.my or trialocsp.digicert.com.my
PREQ_ECONN_005	Allows connectivity from DSS to QR Proxy [port 443] Production is [will provide later in a separate document] Staging is dev-bb8.posdigicert.com.my
PREQ_ECONN_006	Allows connectivity from QR Proxy to document server [port 443] Production is [will provide later in a separate document] Staging is dev-bb8.posdigicert.com.my *__Note__: Document server should have valid SSL Document server could only accept in-bound connection from QR Proxy
PREQ_ECONN_007	Allows connectivity from System to MYCRS API [port 7443] Production is [will provide later in a separate document] Staging is dev-mycrs.posdigicert.com.my
PREQ_ECONN_008	Allows connectivity to MYCRS AP [port 443] Production is [will provide later in a separate document] Staging is demo-mycrs.posdigicert.com.my

Table 4: Pre-requisite: External Connection

5 Digital Certificate Request and Issuance

5.1 Digital Certificate Information

Digital Certificate information is as below,

ITEMS	REMARKS
Medium	Roaming Certificate for Individual
Type	AATL
Validity	1 year validity **Note*:** *The certificate validity is starting from the date of certificate issuance (date from) and will be valid for 1 year from that date.* *Example: Certificate Expiry Date is on 01/03/2025, the certificate will be valid until 01/03/2026. After the 1 year period, certificate need to be renewed*
Charging Method	**Subscription based (unlimited signing)**
Supporting Document	• Given during certificate request via API 1. Copy of IC (front & back) for Malaysian / Passport for Non-Malaysian **Note**:** d) IHH will provide a master LOA for all certificate requests. LOA is not required for subsequent certificate requests. e) For certificate revocation process, IHH will also only use the master LOA. New LOA is not required for any certificate revocation request as per IHH's letter of responsibility.
DN String Structure	Information that will appear in digital certificate's subject attribute are as below: • SERIALNUMBER = <NRIC number / Passport No> (Please omit dash for NRIC number) • CN = <Name> (As per NRIC / Passport) • O = <Company Name> • OU = <Company Registration No.> (Please use the new format from SSM) • E = <Email Address> • C = <Citizenship> ***Note:** 1) O and OU will refer to the company where he/she works (if applicable)
Certificate Request & Issuance Method	• Verified by IHH AP via System • Immediate Issuance (Trusted Database) ○ IHH must provide a letter assuming full responsibility for certificate issuance and revocation request to Pos Digicert, as advised by Pos Digicert's legal team • IHH able to do Bulk or Ad-Hoc certificate request but subject to **PREQ_SYSTEM_004** **Note**:** IHH will be able to renew certificate at any time

5.2 Digital Certificate Issuance (Trusted Database)

As per **Figure 2,** IHH Admin will do Digital Certificate issuance process.

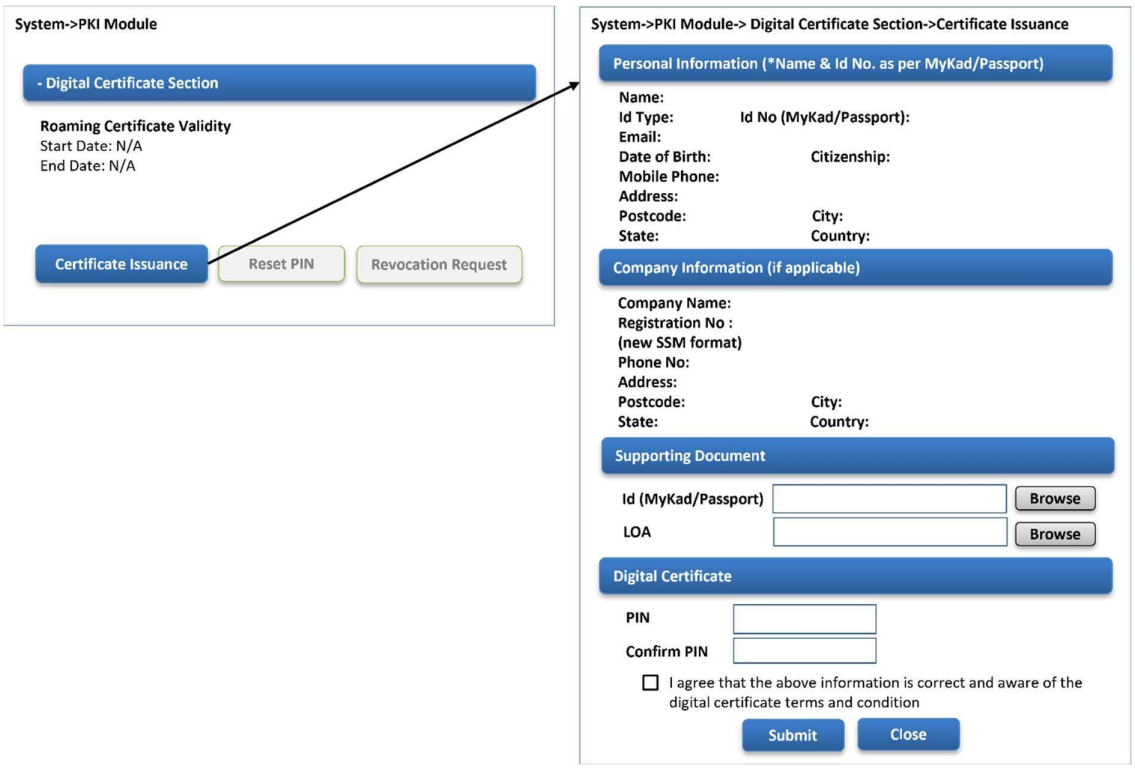


Illustration 1: Digital Certificate Request and Issuance

1) System will display digital certificate request form for IHH Admin to verify and enter. System can prepopulate available information from the System. System must make sure that
   a) Identity No. and Name as per supporting document. Please omit symbol dash in MyKad
   b) IHH Admin is aware on the digital certificate issuance
   c) Save/cache PIN entered as per IHH implementation

   Note:** For Bulk registration, System will be responsible to handle the process accordingly

2) System will perform digital certificate issuance by using MYCRS API
   a) Please save returned information from MYCRS API. MYCRS Request Code is needed for revocation request
   b) System could send email notification to IHH Admin on the successful of digital certificate issuance and include information on next action.
   c) System must make sure Identification No., Name, Organisation Name and Organisation Registration No. remains as per digital certificate issuance

5.2.1 Digital Certificate Renewal

1) Information regarding MYCRS and Digital Certificate Renewal,
   a) MYCRS will not send email notification to User upon digital certificate request and issuance
   b) As per current standard practice, the rules are as follows:
      Rule #1: MYCRS API will check existing certificates and will only allow for renewal within 1 month
      Rule #2: MYCRS API will not allow certificate generation if **Rule #1** is not met.

      Due to IHH implementation for renewal at any time, there will be changes on MYCRS API where MYCRS API won't check for **Rule #1** and **Rule #2**. MYCRS API will allow certificates to be generated at any time upon renewal with 1 year validity.

      Example: *Certificate created on 01/01/2025 and has validity until 01/01/2026. If the certificate is renewed on 01/06/2025, the expiry date will be 01/06/2026.*

      Note:** IHH System **must make sure** that there will be **no duplication** of digital certificate.

   c) For IHH implementation, during renewal, no new copy of IC / Passport is required. IHH System to pass existing record of IC/Passport accordingly.
      Note:** IHH System must make sure that the IC/Passport submitted is correct.

   d) Expiry email and renewal of digital certificate is based on send notice parameter,
      i) If set to 1, MYCRS System will NOT send expiry notice to User
      ii) If set to 0, MYCRS System will send expiry notice (30 days, 14 days and 7 days before expiry)

5.2.2 Reset PIN and Revocation Request

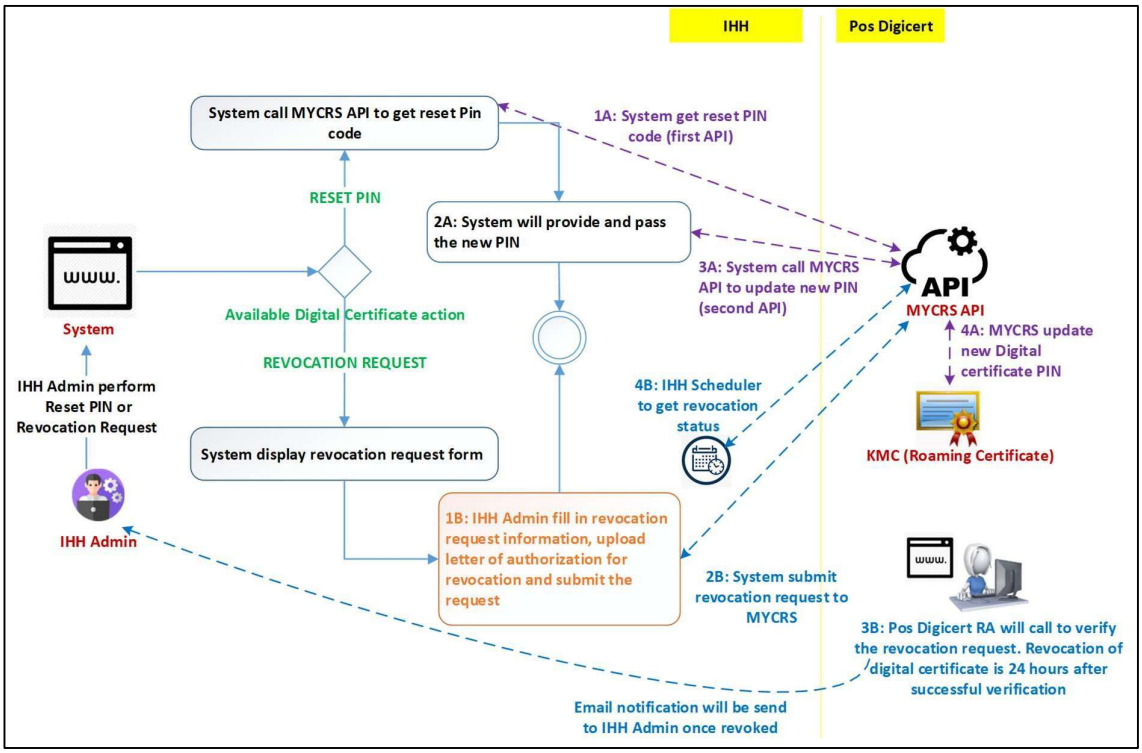


Figure 4: Digital Certificate Reset PIN and Revocation Request

1) System will allow IHH Admin to perform Reset PIN and Revocation Request if Doctor's digital certificate is active

2) When IHH Admin performs **Reset PIN** action,
   a) System will get reset PIN code using MYCRS API
   b) System will provide and pass the new PIN using MYCRS API
   c) System will perform reset PIN by using MYCRS API and update return result. For IHH implementation, entered PIN number will be save/cache by System.

Note:** For IHH implementation,

   1) Reset PIN action will set the new PIN to IHH digital certificate only and will not sync to all available certificates own; and
   2) Reset PIN action will set the new PIN via API only. Should IHH Admin/Doctor reset PIN at MYCRS website, it will not reset the PIN at System.

3) When IHH Admin performs **Revocation Request** action,
   a) System will display revocation form
   b) IHH Admin fills in information, uploads supporting documents (Letter of Authorisation for revocation) and submits the request. System will perform revocation request by using MYCRS API and update return result

Note**:

For revocation, IHH will only use the master LOA provided during certificate request. New LOA is not required for certificate revocation request as per stated in IHH letter of responsibility.

c) Pos Digicert RA will process the request and perform **24 hours revocation SLA** after successfully verify the request. MYCRS will send email on revocation notification to IHH Admin

d) IHH Admin could update revocation status at System via MYCRS API

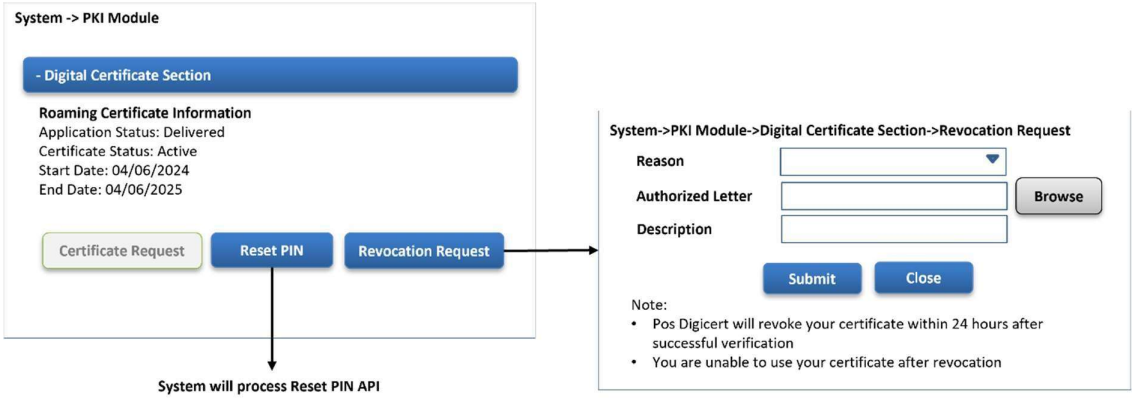


Illustration 2: Reset PIN and Revocation Request

5.3 List of MYCRS API's

This section will describe immediate issuance API for Individual certificate request and issuance. Available APIs are,

- **EnrollRoamingCert** on **page 24**
  - To register Doctor information, certificate request and issuance by IHH Admin in MYCRS
- **GenerateResetPinCode** on **page 27**
  - First API for reset PIN action
- **ResetPin** on **page 28**
  - Second API for reset PIN action
- **RequestRevoke** on **page 29**
  - For revocation request. This request will need approval from Pos Digicert RA
- **certificateStatusIIH** on **page 31**
  - To get revocation status

5.3.1 EnrollRoamingCert

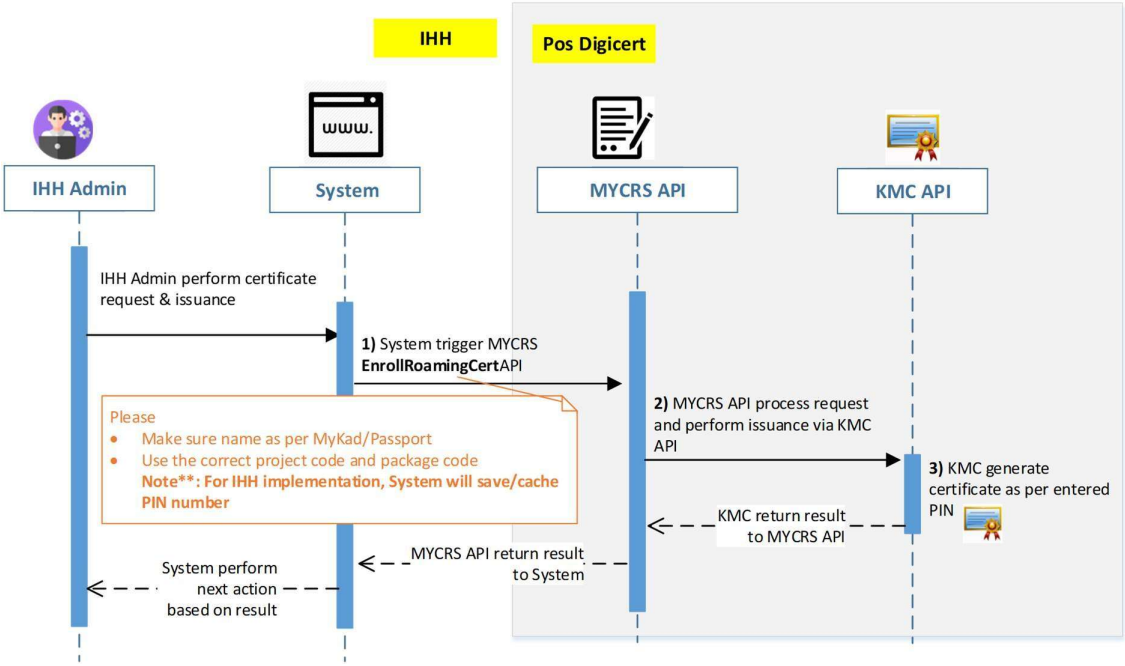


Figure 5: Immediate Certificate Issuance Sequence Diagram

PROTOCOL	HTTPS
MESSAGE FORMAT	SOAP/XML
URL (Staging)	
URL (Production)	
FUNCTION	**EnrollRoamingCert**

PURPOSE	To register User information and certificate issuance in MYCRS. **Note for renewal:** Certificate renewal may be done at any time

5.3.1.1 Request Parameter

REQUEST				
NAME		**TYPE**	**LENGTH**	**DESCRIPTION / RULE**
projectCodesetProfile		String	50	Mandatory. MYCRS project code. Will be provided once provisioned by MYCRS. Value for Staging and Production is different
productCode		String	20	Mandatory. MYCRS package code. Will be provided once provisioned by MYCRS. Value for Staging and Production is different
userProfile		userProfile		Please refer **userProfile** structure below
1	Name	String	64	Mandatory. As per IC or Passport
2	Idtype	String	2	Mandatory. Default to “IC” for IC and “PP” for passport
3	Idno	String	15	Mandatory. IC (Without “-“) or Passport No.
4	Email	String	80	Mandatory
5	Dob	String	10	Date of birth. Format is DD-MM-YYYY
6	Citizenship	String	3	Mandatory. Please refer Country and State Code list on **page 32**
7	Telmobile	String	50	Please set null value to blank if not available
8	Telhome	String	50	Please set null value to blank if not available
9	Faxno	String	50	Please set null value to blank if not available
10	Add	String	250	Mandatory. Combine all address in one field
11	Postcode	String	50	Mandatory
12	City	String	50	Mandatory. Free text
13	State	String	3	Mandatory. Please refer Country and State Code list on **page 32**
14	Country	String	3	Mandatory. Please refer Country and State Code list
15	BussinessType	String	1	Mandatory. “I” for individual, “C” for Company. If you set as “I”, API will auto set Individual information as Company information, even though you send the Company information
16	CompName	String	64	Mandatory if BussinessType is “C”
17	CompRegNo	String	64	Mandatory if BussinessType is “C”
18	Comptel	String	50	Please set null value to blank if not available
19	CompFaxno	String	50	Please set null value to blank if not available
20	CompAdd	String	250	Mandatory if BussinessType is “C”. Combine all addresses line in one field.
21	CompCity	String	50	Mandatory if BussinessType is “C”. Free text
22	CompState	String	3	Mandatory if BussinessType is “C”. Please refer Country and State Code list on **page 32**
23	CompCountry	String	3	Mandatory if BussinessType is “C”. Please refer Country and State Code list on **page 32**
24	CompPostcode	String	50	Mandatory if BussinessType is “C”
document		document		Please refer **document** structure below
1	Key	String		Mandatory.

REQUEST				
NAME		TYPE	LENGTH	DESCRIPTION / RULE
				• For ID: Please set as "NRIC/Passport" • For LOA: Please set as "Authorization Letter"
2	FileType	String	10	Mandatory. Available values are PDF and JPEG
3	Image	byte		Mandatory
pin		String	16	Mandatory. Certificate password. PIN length must be 8-16 alphanumeric character, at least 1 capital letter, at least 1 small letter, at least 1 number and at least 1 symbol
sendNotice		String	1	Please set to '1'. [1 – do not send expiry notice, 0- send expiry notice]

5.3.1.2 Response Parameter and Status Code

RESPONSE			
NAME	TYPE	LENGTH	DESCRIPTION / RULE
requestCode	String	50	MYCRS unique request code. Please keep this value for reference. Use in revocation request
statusCode	String	4	-Refer below-
statusMessage	String	100	-Refer below-
CertinBased64			Certificate for information only
ValidFrom	String	30	Format is yyyy-mm-dd hh:mm:ss ex.2021-02-05 14:30:43
ValidTo	String	30	Format is yyyy-mm-dd hh:mm:ss ex.2021-02-12 14:30:43

STATUS CODE & MESSAGE	
CODE	MESSAGE
9001	Success
9003	Incomplete Parameter
	Incomplete Parameter: Supporting document NULL
	Invalid email format
	Invalid IC Format
	Incorrect business type
9004	Invalid Product Code
9005	Invalid Project Code
9012	PIN length must be 8-16 alphanumeric character
9009	General Error
9015	Wrong file type
9016	MYCRS Request Code does not exist

5.3.2 GenerateResetPinCode

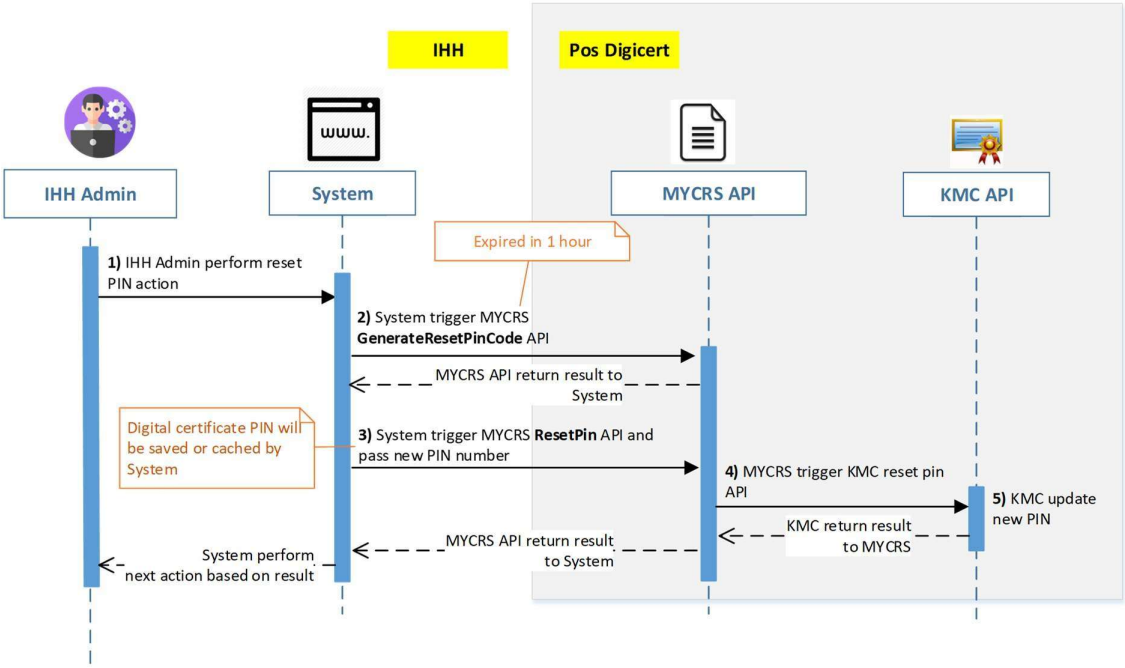


Figure 6: Reset Pin Sequence Diagram (Direct)

PROTOCOL	HTTPS
MESSAGE FORMAT	SOAP/XML
URL (Staging)	
URL (Production)	
FUNCTION	**GenerateResetPinCode**
PURPOSE	To get activation code for reset roaming certificate PIN / password. (First API)

5.3.2.1 Request Parameter

REQUEST			
NAME	**TYPE**	**LENGTH**	**DESCRIPTION / RULE**
projectCodesetProfile	String	50	Mandatory. Will be provided once provisioned by MYCRS. Value for Staging and Production is different
userId	String	15	Mandatory. Please set as Identification No. Same value as certificate request
email	String	100	Mandatory. Email when creating the certificate. Same value as certificate request

5.3.2.2 Response Parameter and Status Code

RESPONSE			
NAME	**TYPE**	**LENGTH**	**DESCRIPTION / RULE**
resetCode	String	20	Reset PIN activation code.

RESPONSE			
NAME	**TYPE**	**LENGTH**	**DESCRIPTION / RULE**
statusCode	String	4	-Refer below-
statusMessage	String	100	-Refer below-

STATUS CODE & MESSAGE	
CODE	**MESSAGE**
9001	Success
9003	Incomplete Parameter
9005	Invalid Project Code
9993	Certificate not found
9011	Email not match
9009	General Error
9031	Email address is not valid

5.3.3 ResetPin

Please refer diagram at **GenerateResetPinCode** on **page 27**.

PROTOCOL	HTTPS
MESSAGE FORMAT	SOAP/XML
URL (Staging)	
URL (Production)	
FUNCTION	**ResetPin**
PURPOSE	To reset roaming certificate PIN / password.

5.3.3.1 Request Parameter

REQUEST			
NAME	**TYPE**	**LENGTH**	**DESCRIPTION / RULE**
projectCodesetProfile	String	50	Mandatory. Will be provided once provisioned by MYCRS. Value for Staging and Production is different
userId	String	15	Mandatory. Please set as Identification No. Same value as certificate request
resetCode	String	20	Mandatory. Activation code from **GenerateResetPinCode** API
newPIN	String	16	Mandatory. Certificate password. PIN length must be 8-16 alphanumeric character, at least 1 capital letter, at least 1 small letter, at least 1 number and at least 1 symbol

5.3.3.2 Response Parameter and Status Code

RESPONSE			
NAME	**TYPE**	**LENGTH**	**DESCRIPTION / RULE**
statusCode	String	4	-Refer below-
statusMessage	String	100	-Refer below-
STATUS CODE & MESSAGE			

RESPONSE			
NAME	TYPE	LENGTH	DESCRIPTION / RULE
CODE	MESSAGE		
9001	Success		
9003	Incomplete Parameter		
9012	PIN length must be 8-16 alphanumeric character		
9009	General Error		
9008	Activation Code not match		

5.3.4 RequestRevoke

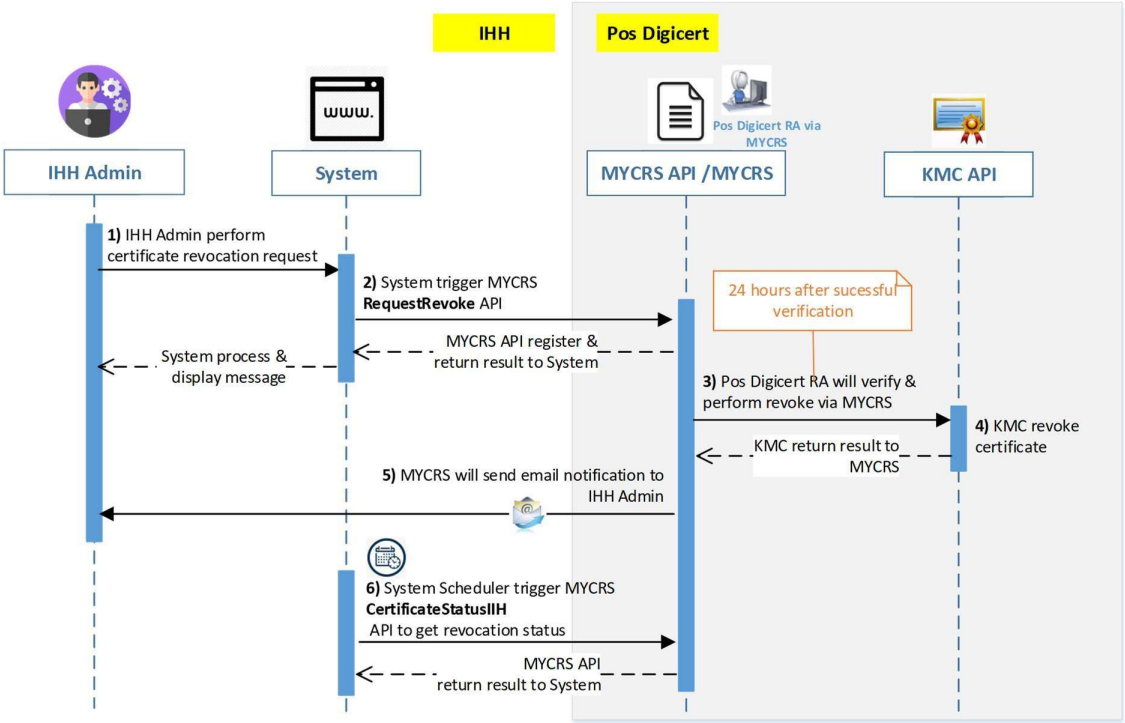


Figure 7: Revocation Request Sequence Diagram

PROTOCOL	HTTPS
MESSAGE FORMAT	SOAP/XML
URL (Staging)	
URL (Production)	
FUNCTION	**RequestRevoke**
PURPOSE	Request to revoke/terminate the certificate

5.3.4.2 Request Parameter

Note:** As per IHH's letter of responsibility on revoke process,

1) IHH is required to send a blank PDF as a default value for any revoke request.
2) LOA required during revoke request will be referred to the master LOA given by IHH during certificate request

REQUEST			
NAME	**TYPE**	**LENGTH**	**DESCRIPTION / RULE**
projectCode	String	50	Mandatory. Will be provided once provisioned by MYCRS. Value for Staging and Production is different
userId	String	15	Mandatory. Please set as Identification No. Same value as certificate request
requestnCode	String	50	Mandatory. MYCRS Request code from **EnrollRoamingCert** API
reasonCode	String	16	Mandatory. Available values are [1, 5, -2] 1: KEY_COMPROMISED (Compromised) 5: CESSATION_OF_OPERATION (Termination) -2: NO_REASON_CODE (Other)
reasonDesc	String	20	Revocation reason. Mandatory for Other
docType	String	10	Mandatory. Available values are PDF or JPEG
authLetter	String	Unlimited based on document	Mandatory. Source file for supporting document in Based64 format

5.3.4.3 Response Parameter and Status Code

RESPONSE			
NAME	**TYPE**	**LENGTH**	**DESCRIPTION / RULE**
statusCode	String	4	-Refer below-
statusMessage	String	100	-Refer below-

STATUS CODE & MESSAGE	
CODE	**MESSAGE**
9001	Success
9003	Incomplete Parameter
	Revoke Request Is In Processing
	Incomplete Parameter, reasonCode =-2, reason description is null
9007	Invalid User Id
9009	General Error
9015	Wrong file type
9016	Request Id not found
9017	Certificate not found
9018	Certificate already expired
9019	Certiifcate already revoke
9020	Success. But unable to send email notification to RA
9030	Please revoke by using this Request Code: {LatestRequestCode} instead

5.3.5 certificateStatusIIH

Usage as per RequestRevoke diagram on **page 29**.

PROTOCOL	HTTPS
MESSAGE FORMAT	SOAP/XML
URL (Staging)	
URL (Production)	
FUNCTION	**certificateStatusIIH**
PURPOSE	To get certificate information and status based on given ID.

5.3.5.1 Request Parameter

REQUEST			
NAME	**TYPE**	**LENGTH**	**DESCRIPTION / RULE**
Sender	String	20	Mandatory. MYCRS project code. Will be provided once provisioned by MYCRS. Value for Staging and Production is different.
IdNo	String	50	Mandatory. Value same as certificate request.

5.3.5.2 Response Parameter and Status Code

RESPONSE				
NAME		**TYPE**	**LENGTH**	**DESCRIPTION / RULE**
Cert		CertList	N/A	Please refer CertList Structure as below
1	CodeRequest	String	50	MYCRS Request Code from **EnrollRoamingCert** API
2	IcNo	String	15	IC no as per request
3	Status	String	10	Valid values are 0,1,**2**,3,4,5 0 - PROCESSING, 1-ACTIVE, **2 - REVOKED**, 3 – EXPIRED, 4 – INACTIVE and 5-CLOSED
4	TraderId	String	50	Company Registration No. as per request
5	ValidFrom	String	10	Format "DD-MM-YYYY"
6	ValidTo	String	10	Format "DD-MM-YYYY"
Recordcount		String	4	
Statuscode		String	N/A	-Refer below-

STATUS CODE	
001	Record Exists
002	Record Not Exists

5.4 List of Country Code and State Code

Please refer to List of Country Code and State Code. Citizenship value is the same as the Country Code.

List of country and state code other than Malaysia can be retrieved based on ISO 2-letter country code.

COUNTRY NAME / CITIZENSHIP	COUNTRY /CITIZENSHIP CODE (VALUE TO PASS)	STATE NAME	STATE CODE (VALUE TO PASS)
Malaysia	MY	Johor	01
Malaysia	MY	Kedah	02
Malaysia	MY	Kelantan	03
Malaysia	MY	Melaka	04
Malaysia	MY	Negeri Sembilan	05
Malaysia	MY	Pahang	06
Malaysia	MY	Perak	07
Malaysia	MY	Perlis	08
Malaysia	MY	Pulau Pinang	09
Malaysia	MY	Selangor	10
Malaysia	MY	Terengganu	11
Malaysia	MY	Sabah	12
Malaysia	MY	Sarawak	13
Malaysia	MY	Federal Territory of Kuala Lumpur	14
Malaysia	MY	Federal Territory of Labuan	15
Malaysia	MY	Federal Territory of Putrajaya	16

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6 Authorised Personnel (AP)

IHH must become Pos Digicert Authorised Personnel (AP). Requirements to be an AP are as below:

a) Must be at least on Executive / Officer level of the organization
b) Nomination must be made by the organization
c) Complete and attached together the following
   i) Copy of NRIC (font and back) / Passport of the nominated AP
   ii) Nomination Letter from the organization
   iii) Application Form. Please use the link below to download the form.
      (1) https://www.posdigicert.com.my/public/uploads/forms/Authorised_Personnel_AP_Form.pdf
   iv) Payment (Cheque / Bank Slip / Postal Order / Purchase Order) (if applicable)
d) Please fill in form and pass the request to Project Manager / Account Manager

Pos Digicert Registration Authority (RA) will process the application accordingly

- Once approved, Authorized Personal (AP) will receive an email notification to set Roaming Certificate PIN. Please set your PIN accordingly
- To view certificate request or to request revoke,
  - Browse to https://mycrs.posdigicert.com.my/home/ap
  - The credential will be your IC No. and PIN

7 Digital Signing Solution (DSS)

This section will describe API for PDF Signing.

7.1 Scenarios and Highlights

Below are the scenarios and highlights of PDF Signing using DSS.

****Note:** Generation of QR code is optional for implementation. It is not mandatory at this stage but may serve as a reference for potential future deliverables.

SCENARIO	INDIVIDUAL		
	Signing	Remarks	QR Code
Scenario 1	√		
Scenario 2	√	√	
Scenario 3	√		√
Scenario 4	√	√	√

Table 5: PDF Signing Scenarios

1) Generation of QR Code must be initiated during first signing and it is controlled by System. Please pass the signed pdf for subsequence signer
2) Document validity is controlled by System
3) Filename for source and destination must be unique to avoid system referring to a wrong document
4) The creation of signature image file (if used) is under System responsibility. Please use signature image, if document is applicable for stamping

Illustration 3: PDF Signing

Note: Example instruction "Scan to Verify" is part of PDF content.

7.2 Document Verification (QR Code) via eValidator

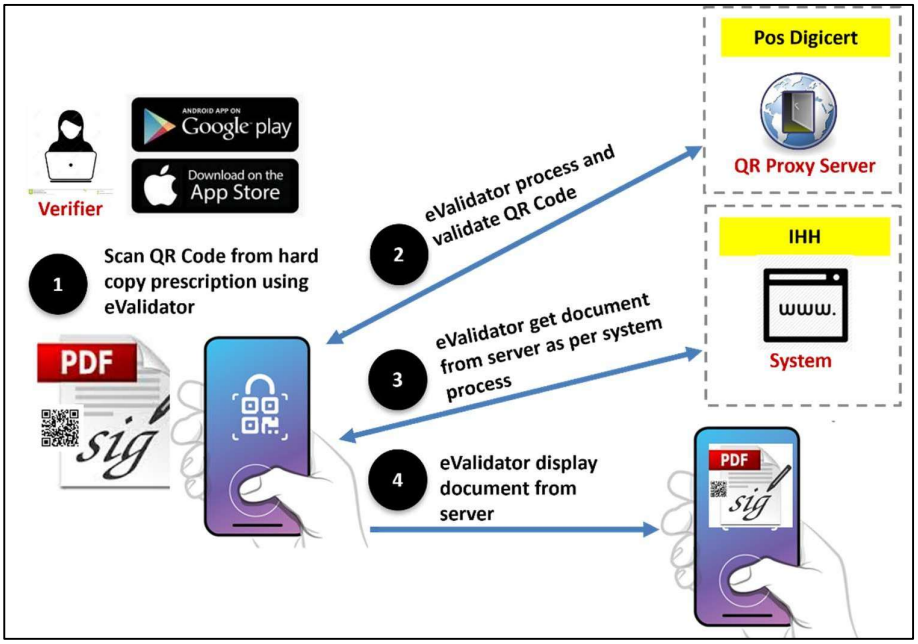

Figure 8: Document Verification (QR Code) via eValidator

***Note:** Document verification (QR Code) via eValidator is optional for implementation. It is not mandatory at this stage but may serve as a reference for potential future deliverables.

1) Purpose of eValidator Mobile Application:
   a) Will verify and validate the authenticity of physical document. Available in Google Play (android) and App Store (iOS)

2) Used by Verifier / Internal User to verify authenticity of physical document:
   a) Install eValidator from Google Play (Android) or App Store (iOS) on smartphone
   b) Run the application
   c) Press button and point the smartphone camera to the QR Code
   d) eValidator will display the original copy of the document stored in Digital Signing Registration system server that has been digitally signed
   e) Ensure that the copy appeared in the smartphone matches the printed copy

***Note:** System could provide guidelines of how to install and use eValidator as part of PDF document for easier reference by Verifier / Internal User.

7.3 List of DSS API's

Available API are,

- **verifyRoamingCert** on **page 39**
  - To check on certificate status and existence
- **verifyRoamingPin** on **page 41**
  - To verify roaming pin before proceeding with signing
- **signPdfConfigB** on **page 43**
  - To sign pdf file using roaming certificate

Note: Validating the roaming certificate status and PIN (password) before signing is good practice. This is to make sure that the signing process is done on a valid certificate and a correct PIN (password).

Please refer to the next page.

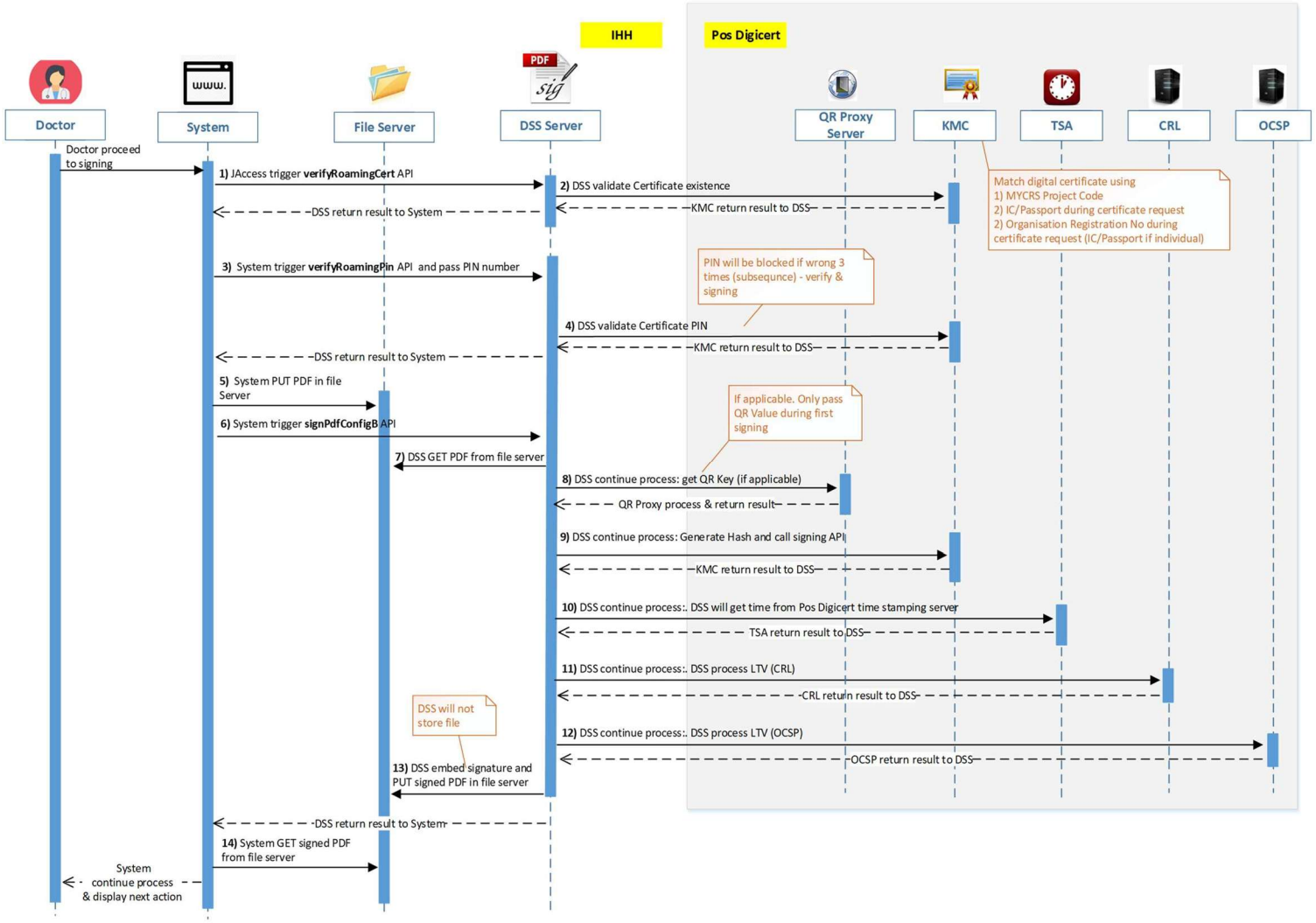

Figure 9: PDF Signing Sequence Diagram

Figure 9 describes the details process for Individual PDF Signing. When Doctor performs signing action at System, System will not proceed to the next step if error occur and will prompt message accordingly.

1) System **validates digital certificate status** using DSS API. System will proceed to respective action based on return status. For digital certificate matching, please make sure System pass the correct value as per digital certificate request and issuance
    i) For Individual with Company: **IC/Passport No.**, **Company Registration No.** and **MYCRS Project Code**
    ii) For Individual without Company: **IC/Passport No.** and **MYCRS Project Code**. (When calling API, please pass IC/Passport No. value in **Company Registration No. field**)

2) System will pass PIN number and **validate digital certificate PIN** using DSS API and proceed if matched.
    a) System will PUT uploaded PDF file to file server and call DSS API
    b) DSS will start signing process
        i) DSS will GET PDF file from file server
        ii) DSS will generate QR Code (if applicable) and proceed to perform signing
        iii) DSS will proceed with others signing related process that are Time Stamping and LTV
            (1) Time from TSA will appear in ADOBE signature properties
            (2) Time display in the PDF is from DSS server
        iv) DSS will PUT signed PDF in file server and return result to System
    c) System will GET signed PDF and proceed with applicable action at System
    d) System will display signing process result to Doctor accordingly

7.3.1 verifyRoamingCert

PROTOCOL	HTTPS / POST / x-www-form-urlencoded
MESSAGE FORMAT	RESTFUL
URL (Staging)	
URL (Production)	
MEDIATYPE	application/json
FUNCTION	**verifyRoamingCert**
PURPOSE	To check roaming certificate existence. Certificate is Revoked and Certificate is Expired are same as certificate not available.

7.3.1.1 Request Parameter

REQUEST			
NAME	**TYPE**	**LENGTH**	**DESCRIPTION / RULE**
pCode	String	45	Mandatory. MYCRS Project Code. Will be provided once provisioned by MYCRS. Value for Staging and Production is different.
userId	String	20	Mandatory. Identification Number. IC or Passport when applying for the roaming certificate.
orgId	String	45	Mandatory. Value set when applying the roaming certificate.

7.3.1.2 Response Parameter and Status Code

RESPONSE		
NAME	**TYPE**	**DESCRIPTION / RULE**
statusCode	String	-Refer below-
certificate	String	Certificate details for information only

STATUS CODE	
901	Success
902	Invalid Parameter
903	Internal Server Error
800	Operation Roaming Failed
802	PIN is Blocked
805	User Not Exist
808	Certificate is Revoked
811	Certificate is Expired

7.3.1.3 Sample Code (JAVA)

```
package com.posdigicert.test;

import  java.io.IOException;

import javax.ws.rs.core.MediaType;

import javax.ws.rs.core.MultivaluedMap;

import javax.ws.rs.core.UriBuilder;

import com.sun.jersey.api.client.Client;

import com.sun.jersey.api.client.ClientResponse;
```

```
import com.sun.jersey.api.client.WebResource;

import com.sun.jersey.core.util.MultivaluedMapImpl;


public class VerifyCert {

  public static void main(String[] args) throws IOException {

    // TODO Auto-generated method stub

    VerifyCert.verify3();

  }

  private static void verify3() {

    try {

      Client client = Client.create();

      WebResource webResource = client

         .resource(UriBuilder.fromUri("https://demo-
kit.posdigicert.com.my/DssManagerApi/rest/roaming/verifyRoamingCert").build());

      MultivaluedMap<String, String> formData = new MultivaluedMapImpl();

     formData.add("pCode", "as per provide code");

     formData.add("userId", "as per sample");

      formData.add("orgId", "as per sample");

      ClientResponse response = webResource.type(MediaType.APPLICATION_FORM_URLENCODED_TYPE)

         .post(ClientResponse.class, formData);

      System.out.println("Output from Server……\n");

      String output = response.getEntity(String.class);

      System.out.println(output);

    } catch (Exception e) {

      e.printStackTrace();

    }

  }

}
```

7.3.2 verifyRoamingPin

PROTOCOL	HTTPS / POST / x-www-form-urlencoded
MESSAGE FORMAT	RESTFUL
URL (Staging)	
URL (Production)	
MEDIATYPE	application/json
FUNCTION	**verifyRoamingPin**
PURPOSE	To verify roaming certificate PIN before proceeding with signing.

7.3.2.1 Request Parameter

REQUEST			
NAME	**TYPE**	**LENGTH**	**DESCRIPTION / RULE**
pCode	String	45	Mandatory. MYCRS Project Code. Will be provided once provisioned by MYCRS. Value for Staging and Production is different.
userId	String	20	Mandatory. Identification Number. IC or Passport when applying for the roaming certificate.
orgId	String	45	Mandatory. Organization Reference. Company Name, Registration No. or Identification No. when applying for the roaming certificate.
pin	String	16	Mandatory. The certificate PIN (password).

7.3.2.2 Response Parameter and Status Code

RESPONSE		
NAME	**TYPE**	**DESCRIPTION / RULE**
statusCode	String	-Refer below-
statusMessage	String	Status message.

STATUS CODE	
901	Success
902	Invalid Parameter
800	Operation Roaming Failed
802	PIN is Blocked
804	PIN Incorrect
805	User Not Exist
808	Certificate is Revoked

7.3.2.3 Sample Code (JAVA)

```
package com.posdigicert.test;

import javax.ws.rs.core.MediaType;

import javax.ws.rs.core.MultivaluedMap;

import javax.ws.rs.core.UriBuilder;

import com.sun.jersey.api.client.Client;

import com.sun.jersey.api.client.ClientResponse;

import com.sun.jersey.api.client.WebResource;
```

```
import com.sun.jersey.core.util.MultivaluedMapImpl;


public class VerifyRoamingPin {

  public static void main(String[] args) {

    VerifyRoamingPin.verifyPin();

  }

  private static void verifyPin() {

    try {

      Client client = Client.create();

      WebResource webResource = client

         .resource(UriBuilder.fromUri("https://demo-
kit.posdigicert.com.my/DssManagerApi/rest/roaming/verifyRoamingPin").build());

      MultivaluedMap<String, String> formData = new MultivaluedMapImpl();

     formData.add("pCode", "as per provide code");

     formData.add("userId", "as per sample");

      formData.add("orgId", "as per sample");

      formData.add("pin", "as per sample");//please use actual PIN that user set

      ClientResponse response = webResource.type(MediaType.APPLICATION_FORM_URLENCODED_TYPE)

         .post(ClientResponse.class, formData);

      System.out.println("response.getStatus() : "+response.getStatus());

      System.out.println("Output from Server……\n");

      String output = response.getEntity(String.class);

      System.out.println(output);

    } catch (Exception e) {

      e.printStackTrace();

    }

  }

}
```

7.3.3 signPdfConfigB

PROTOCOL	HTTPS / POST / x-www-form-urlencoded
MESSAGE FORMAT	RESTFUL
URL (Staging)	
URL (Production)	
MEDIATYPE	application/json
FUNCTION	**signPdfConfigB**
PURPOSE	To sign PDF using roaming individual certificate.

7.3.3.1 Request Parameter

****Note:**

1) Filename for **source** and **dest** must be **unique** to avoid system referring to wrong document.

2) If QR Code is **applicable**, please provide **only on first signer signing action (one time)**. Please install eValidator for QR Code verification. eValidator is built by Pos Digicert and is available at Apps and Play Store.

REQUEST			
NAME	**TYPE**	**LENGTH**	**DESCRIPTION / RULE**
pCode	String	45	Mandatory. MYCRS Project Code. Will be provided once provisioned by MYCRS
source	String	300	Mandatory. PDF file path and name to be signed. Please do not set mapped path of the file server Ex: • Map drive of file server is “D:/CTC”. • Folder is “Input”. • File name is “PDFDocNotSign.pdf”. • **source is “/Input/ PDFDocNotSign.pdf”**
dest	String	300	Mandatory. PDF file path and name after signed. Please do not set mapped path of the file server Ex: • Map drive of file server is “D:/CTC”. • Folder is “Output”. • File name is “PDFDocSigned.pdf”. • **dest is “/Output/ PDFDocSigned.pdf”**

signerXyPage	String	300	Mandatory. Signing position and page number for signature to appear. Signer XY coordinate & signing page(s)
			• Comma separated string, that consist of below information in sequence:
			  ○ XStart
			  ○ YStart
			  ○ XEnd
			  ○ YEnd
			  ○ Page(s)
			• Ex: 100,200,50,25,2 (page 2 only)
			• Ex: 100,200,50,25,2,3,4 (page 2,3 & 4)

REQUEST			
NAME	**TYPE**	**LENGTH**	**DESCRIPTION / RULE**
			<u>If signing location is different</u>: Example: 1) Location and page: 10,10,50,50,1,2,3 2) Location and page: 100,100,150,150,1,3,5 3) Location and page: 200,200,250,250,1,2,4 Please separate location and page using "I" pipe symbol as below: • 10,10,50,50,1,2,3|100,100,150,150,1,3,5|200, 200,250,250,1,2,4 **Note:** Number of times Signature will appear in ADOBE signature list will be based on number of different Location and page
signerImagePath	String	300	Optional. Signer's signature image path and file name. Please do not set mapped path of the file server Ex: • Map drive of file server is "D:/CTC". • Folder is "SignatureImage". • File name is "901126025076.png". • **signerImagePath is "/SignatureImage/901126025076.png"**
qrValue	String	200	Optional. String value that will be generated to encrypted QR code. The encrypted QR code will appear in PDF
qrPosition	String	300	Mandatory if qrValue is provided. QR image coordinates and page number(s). • Comma separated String, that consist of below information in sequence: i. Absolute path X ii. Absolute path Y iii. QR page(s) iv. Ex: 50,50,1 (page 1 only) Ex:50,50,2,3,4 (page 2,3 & 4)
qrSize	Integer	3	Mandatory if qrValue is provided. Allocation Size for QR Code.
signerFontSize	Integer	3	Optional. Signature Text Font Size. Ex: "Digitally Signed by …."
customSignatureText	String	80	Optional. To replace "Digitally signed by .." and "Digitally signed on .." wording Example: • Not applicable if using standard wording

REQUEST			
NAME	**TYPE**	**LENGTH**	**DESCRIPTION / RULE**
			• Senarios for changes as below **Senarios** / **Example Rules** Replace all Words — Tandatangan Oleh– Tandatangan Pada Replace "Digitally signed by .." words only — Tandatangan Oleh– Replace "Digitally signed on .." words only — –Tandatangan Pada Replace and Appear only "Digitally signed by .." words only — Tandatangan Oleh–* Replace and Appear "Digitally signed on .." words only — *–Tandatangan Pada *Note: • Please make sure no spaces before and after dash symbol • To replace Example Rule as per your system requirements
signerTextRow	Integer	3	Optional. Signature Text Row. Signature text start row
dtsFlag	Integer	1	Mandatory. Please default to 1
userId	String	20	Mandatory. Identification Number. IC or Passport when applying for the roaming certificate.
orgId	String	45	Organization Reference. Mandatory. Company Name, Registration No. or Identification No. when applying the roaming certificate
pin	String	16	Mandatory. The certificate PIN (password)
remark	String	300	Optional. Not applicable. Please default to blank
fileServerId	String	20	Mandatory. File Server Id. Will be provided after DSS configuration

7.3.3.2 Response Parameter and Status Code

RESPONSE			
NAME	**TYPE**	**LENGTH**	**DESCRIPTION / RULE**
requestId	String	10	Transaction ID assigned by DSS
statusCode	String	3	-Refer below-

STATUS CODE	
901	Success
902	Invalid Parameter
903	Internal Server Error
904	File Server Access Error
905	Invalid Signing Page **Note: Invalid Signing Page error is refer to given page number for signing does not exist.*

RESPONSE			
NAME	TYPE	LENGTH	DESCRIPTION / RULE
907	Error QR Transaction		
908	File is locked		
912	Error LTV Verification		
913	Error LTV Process		
800	Operation Roaming Failed		
802	PIN is Blocked		
804	PIN Incorrect		
805	User Not Exist		
808	Certificate is Revoked		
811	Certificate is Expired		

7.3.3.3 Sample Code (JAVA)

```
package com.posdigicert.test;

import java.text.SimpleDateFormat;

import java.util.Date;

import javax.ws.rs.core.MediaType;

import javax.ws.rs.core.MultivaluedMap;

import javax.ws.rs.core.UriBuilder;

import com.sun.jersey.api.client.Client;

import com.sun.jersey.api.client.ClientResponse;

import com.sun.jersey.api.client.WebResource;

import com.sun.jersey.core.util.MultivaluedMapImpl;


public class SignRoamingB {

  public static void main(String[] args) {

    SignRoamingB.sign();

  }

  private static void sign() {

    SimpleDateFormat sdf = new SimpleDateFormat("ddMMyyyyHHmmSSS");

    try {

      Client client = Client.create();

      WebResource webResource = client.resource(

         UriBuilder.fromUri("https://demo-
kit.posdigicert.com.my/DssManagerApi/rest/roaming/signPdfConfigB").build());

      MultivaluedMap<String, String> formData = new MultivaluedMapImpl();

      formData.add("pCode", "as per provide code");
```

```
    formData.add("userId", "as per sample");

     formData.add("orgId", "as per sample");

     formData.add("pin", "as per sample");//please use actual PIN that user set

     formData.add("source", "/Original/blank3pages.pdf");

     formData.add("dest", "/Signed/blank3pages_signed_rest_conf_b_" + sdf.format(new Date())
+ ".pdf");

     formData.add("signerXyPage", "10,10,50,50,1");

     formData.add("signerImagePath", "");

     formData.add("qrValue", "");

     formData.add("qrPosition", "");

     formData.add("qrSize", "0");

     formData.add("signerFontSize", "7");

     formData.add("signerTextRow", "5");

    formData.add("customSignatureText", "");

    formData.add("dtsFlag", "1");

     formData.add("remark", "");

     formData.add("fileServerId", "wiseai");

     ClientResponse response = webResource.type(MediaType.APPLICATION_FORM_URLENCODED_TYPE)

        .post(ClientResponse.class, formData);

     System.out.println("response.getStatus() : " + response.getStatus());

     System.out.println("Output from Server……\n");

     String output = response.getEntity(String.class);

     System.out.println(output);

   } catch (Exception e) {

     e.printStackTrace();

   }

  }

}
```

8 Contact Us

Please refer below.

Pos Digicert Counter Operating Hours:	**Monday – Thursday:** 09:00am - 05:00pm **Friday:** 09:00am - 05:00pm**.** Lunch Break at 1:00pm - 02:45pm
Pos Digicert Customer Care Contact Information:	**Operating Hours:** **Monday – Thursday:** 09:00am - 05:00pm. Lunch Break at 1:00pm - 02:00pm **Friday:** 09:00am - 05:00pm**.** Lunch Break at 12:30pm - 02:45pm **Contactable via:** **Email**: customercare@posdigicert.com.my **Support Portal**: https://posdigicertsupport.freshdesk.com/support/tickets/new **Phone**: +603-8800 8008 **24/7 Support is contactable via:** **Email**: customercare@posdigicert.com.my **Support Portal**: https://posdigicertsupport.freshdesk.com/support/tickets/new ***Note**:*** *24/7 Support is subject to purchase*
Website:	https://www.posdigicert.com.my
Others:	• Malaysian Time • Closed on Saturdays, Sundays and Public Holidays

Appendices

Appendix A: Sample Digital Certificate Revoke Request LOA

Your Address
And Letter Head

Manager
Registration Authority Department
Pos Digicert Sdn Bhd
Cyberjaya
63000
Malaysia

Date:

Dear Sir/madam,

RE: REVOCATION REQUEST

This letter is for revocation purpose, I hereby authorize our organizational personnel, Mr/Mrs _____, Ic Number/Passport Number _____. He/She is our Technical/Senior Executive and requested to revoked the certificates immediately.

Your attention towards this matter is highly appreciated.

Thank you.

Chief Executive Officer
(Company Name and Official Stamp)

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